2024 IEEE 3rd International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)
29-30 November 2024, IRAB, Dhaka, Bangladesh

2024 IEEE 3rd International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON) | 979-8-3315-3440-0/24/$31.00 ©2024 IEEE | DOI: 10.1109/RAAICON64172.2024.10928543

An Automated Network Intrusion Detection and Identification System by Synergizing AI Techniques

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***Abstract*—Currently, cybercrimes are a serious threat to computer networks. Researchers have come to view Network Intrusion Detection Systems as a critical defensive component and have worked to develop new methods to detect hostile network activity. However, unknown attacks are difficult to detect on a zero-day basis. Previous studies addressed detection or identification techniques using binary classifications of attacks. In this paper, we introduce a system for the detection and identification of network intrusions that combines the strengths of the misuse and anomaly detection techniques. Our approach integrates a Gaussian Mixture Model, identify attacks resembling normal distributions enhances early detection capability, for anomaly detection at zero day and supervised models for identification of known attacks. After that, we evaluated our system using the CICIDS2017 and CICIDS2018 datasets, focusing on unsupervised learning for initial attack detection and supervised learning for multiclass classification. Our empirical findings highlight superior performance in terms of accuracy and F1 score, underscoring the effectiveness of our proposed system.**

***Index Terms*—Network Security, Intrusion Detection, Intrusion Identification, Zero-Day Attack**

I. Introduction

To address the growing challenges in network security, this work focuses on integrating machine learning (ML) techniques, supervised and unsupervised, with intrusion detection systems (IDS). The necessity of this research arises from the evolving nature of network threats, particularly zero-day attacks, which are increasingly difficult to detect using traditional methods. Existing solutions primarily focus on intrusion detection or classification, but rarely combine these two crucial aspects to provide a comprehensive defense. Our approach aims to solve real-world security problems by leveraging unsupervised learning for anomaly detection and supervised learning for classifying known and emerging threats. The motivation behind this work is the gap in current IDS research, where there is a lack of effective systems that combine unsupervised and supervised methods to improve detection accuracy and classification efficiency in real-time.

Recent studies in network intrusion detection provide valuable insights but reveal key limitations. While some approaches enhance detection through deep learning, they often lack generalization across diverse attack types. For instance, models like CNNs have shown strong classification performance but struggle with high-dimensional data and do not effectively address zero-day attack detection which means unknown attacking data detection. Other research has focused on combining anomaly detection with machine learning for zero-day attacks, but these models often fail to generalize across datasets. Aksu et al. [1] applied supervised learning techniques like SVM, KNN, and Decision Trees to the CICIDS2017 dataset, achieving high accuracy in classifying known threats. However, reliance on labeled data limits the detection of zero-day attacks. Similarly, Vinayakumar et al. [2] used CNNs and variants to improve classification accuracy, but they did not address zero-day classification. Several studies focus on detecting unknown attacks. Kherbache et al. [3] employed an unsupervised K-means clustering strategy for anomaly detection, but the approach lacks mechanisms to classify the anomalies. Xiao et al. [4] proposed a CNN-based intrusion detection model that improved detection accuracy but lacked zero-day attack detection and relied on the outdated KDD-CUP99 dataset. Wang et al. [5] combine BIRCH clustering with Autoencoders for anomaly detection, achieving high accuracy on datasets like KDDCUP99 and CICIDS2017. However, the model lacks zero-day attack detection and depends on traditional datasets, limiting adaptability to new threats. Our research builds on these works by combining supervised and unsupervised learning with deep learning techniques, aiming to improve real-time network intrusion detection and zero-day attack classification, offering a more robust and generalized solution. Table I shows the Comparison of Intrusion Detection Systems (IDS) based on Zero-Day Attack Detection, Classification, Generalization, and Limitations in recent studies. In this paper, we propose a hybrid Intrusion Detection System (IDS) with key contributions as follows:

1) Using an unsupervised learning technique for unknown (zero-day attack) and known anomaly detection.
2) After detecting that anomaly, supervised learning methods are used to classify that anomaly using multi-class classification.
3) In addition, the challenge of generalization in intrusion detection across diverse datasets.

The detailed architecture of our proposed system, along with the specific scenario and algorithms employed, is described in Section II. The performance evaluation of the system is presented in Section III, followed by the discussion of the results and future research directions in Section IV.

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TABLE I: Comparison of Recent Studies on Intrusion Detection System

Reference	Zero-Day Detection	Classification	Limitation
Aksu et al. [1]	No	Binary	KNN accuracy was only 0.577%
Vinaya kumar et al. [2]	No	Muticlass	Used the outdated NSL-KDD dataset
Kherbache et al. [3]	Yes	No	No classification mechanism
Xiao et al. [4]	No	Multiclass	Uses outdated KDD-CUP99 dataset
Wang et al. [5]	Yes	Binary	Lacks multiclass support and feature selection.

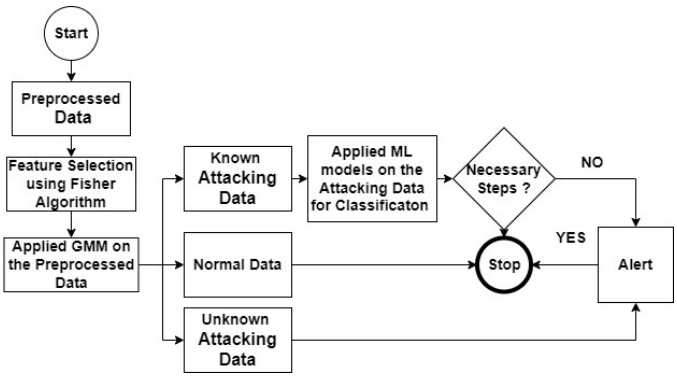


Fig. 1: Diagram of proposed hybrid intrusion detection System

II. Proposed System

In this research, a hybrid IDS system is proposed by combining machine learning with a Gaussian Mixture Model (GMM). Several models, including Decision Tree (DT), K-Nearest Neighbors (KNN), Naive Bayes (NB), and Logistic Regression (LR), were evaluated for performance. The Fisher Score was used for feature selection, and Principal Component Analysis (PCA) reduced GMM processing time. The GMM separates data into normal, known abnormal, and unknown abnormal categories. The machine learning models then classify the known abnormal data, determining which model best distinguishes between anomaly classes. An overview of the entire system is presented in Fig.1.

A. Decision Tree

Decision trees are popular classifiers in machine learning [6], consisting of a root, internal nodes, and leaves that form classification rules. The tree applies these rules until a leaf is reached, where the label determines the classification. Quinlan's C4.5 algorithm [7] selects attributes using information gain.

$$\text{Information Gain} = \text{Entropy}(p) - \sum_{i=1}^{k} \frac{n_i}{n} \text{Entropy}(i) \quad (1)$$

Here, p is the parent node, n_i is the number of records in section i, and Entropy measures the impurity of a node. Higher information gain indicates higher purity. Entropy is computed as:

$$\text{Entropy}(X) = -\sum_{i=1}^{k} P(x=i) \log_2 P(x=i) \quad (2)$$

B. K-Nearest Neighbor (KNN)

The k-Nearest Neighbors (KNN) algorithm is used for both classification and regression tasks. It classifies a new sample by majority vote of its k nearest neighbors in the feature space. The performance of KNN depends on the choice of k and the distance metric, with Euclidean distance being a common choice:

$$d(x_i, x_j) = \sqrt{\sum_{r=1}^{n} (x_{ir} - x_{jr})^2} \quad (3)$$

For classification, KNN assigns the class based on the most frequent class among its neighbors:

$$C(x) = \text{mode}\{C(x_1), C(x_2), \ldots, C(x_k)\} \quad (4)$$

For regression, KNN predicts the value by averaging the values of its neighbors:

$$V(x) = \frac{1}{k} \sum_{i=1}^{k} V(x_i) \quad (5)$$

where $V(x)$ is the predicted value for the new instance [8].

C. Gaussian Naive Bayes

The Gaussian Naive Bayes (GaussianNB) classifier is a probabilistic model for continuous data, based on Bayes' theorem. It assumes normally distributed features, simplifying computations for efficient classification. The likelihood of a feature x given a class c, $P(x|c)$, follows a Gaussian distribution with mean μ_c and variance σ_c^2:

$$P(x|c) = \frac{1}{\sqrt{2\pi\sigma_c^2}} \exp\left(-\frac{(x-\mu_c)^2}{2\sigma_c^2}\right) \quad (6)$$

The parameters μ_c and σ_c^2 are estimated from the training data, and the class probability $P(c)$ is determined by the class frequency in the training set [9].

D. Logistic Regression

Logistic Regression is a statistical method used to model the probability of a binary outcome based on predictor variables (features). It estimates the probability $P(Y=1)$ for a binary outcome Y using the logistic function:

$$P(Y=1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n)}} \quad (7)$$

where $\beta_0, \beta_1, ..., \beta_n$ are model coefficients, estimated via maximum likelihood to fit the observed data [10].

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TABLE II: NEURAL NETWORK STRUCTURE INFORMATION

Layer (type)	Output shape	Activation function	Parameters
Input Layer	(None, 30)	-	0
Dense (fc1)	(None, 64)	ReLU	1984
Dense (fc2)	(None, 32)	ReLU	2080
Dense (fc3)	(None, 24)	ReLU	792
Dense (fc4)	(None, 16)	ReLU	400
Dense (fc5)	(None, 12)	ReLU	204
Dense (fc6)	(None, 8)	ReLU	104
Dense (fcRes)	(None, 8)	Softmax	72

E. Neural Networks

Neural Networks (NNs) are foundational to modern machine learning, modeled after biological neural systems to handle complex patterns. NNs consist of layers of interconnected neurons that process inputs and learn data representations, making them effective in tasks like image recognition and game playing [10]. Our classification model, with 30 input features, includes four hidden layers (128, 64, 48, 32 neurons), ReLU activation, L2 regularization, batch normalization, and dropout (0.5). The output layer has 8 neurons with softmax activation, using sparse categorical crossentropy and the Adam optimizer. Table II shows the structure of the used Neural Network for the analysis to see how it works on the given data.

F. Gaussian Mixture Model

A mixture of distributions for observations is known as a mixture model. Gaussian Mixture Model (GMM) is an algorithm for clustering and density estimation. GMM assumes that the observations are from several Gaussian distributions with unknown parameters, which will be estimated with the expectation maximization (EM) algorithm. GMM models the observation using the following likelihood function [11]:

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x|\mu_k, \Sigma_k) \tag{8}$$

in which the $\mathcal{N}(x|\mu_k, \Sigma_k)$ represents multinomial normal distribution:

$$\mathcal{N}(x|\mu, \Sigma) = \frac{1}{(2\pi)^{\frac{n}{2}} |\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)} \tag{9}$$

When estimating parameters for maximum likelihood (ML) or maximum a posteriori (MAP) in models with unobserved latent variables, the EM approach is applicable. The E-step computes expectations, and the M-step maximizes them; this process is repeated until convergence.

G. Fisher Score

The Fisher Score is a widely used technique for supervised feature selection, reducing dimensionality by identifying the most informative features. It selects the top m features, transforming the input matrix $X \in \mathbb{R}^{d \times n}$ into a reduced matrix $Z \in \mathbb{R}^{m \times n}$.

The Fisher Score is calculated as:

$$F(Z) = \frac{\text{tr}(S_B)}{S_B + cI_{\frac{1}{n}}} \tag{10}$$

where c is a regularization parameter, S_B is the between-class scatter matrix.

For a dataset, the Fisher Score for the j-th feature is:

$$F(x_j) = \frac{\sum_{k=1}^{c} n_k (\mu_k^j - \mu^j)^2}{\sigma_j^2} \tag{11}$$

where c is the number of classes, n_k is the number of samples in class k, and μ_k^j, μ^j, and σ_j^2 represent the means and variance of the j-th feature. Applying the Fisher score, the top m features are selected, reducing dimensionality while retaining the most relevant features.

H. Proposed Hybrid Intrusion Detection System

In our proposed hybrid intrusion detection system (IDS), the pre-processed data, after selecting 30 of the features using Fisher Algorithm which have the higher Fisher Score, is processed through the Gaussian Mixture Model (GMM). The GMM categorizes the data into three types: normal data, known attack data, and unknown data. For normal data, the system allows it to pass through without triggering any alerts. In the case of unknown attack data, the system generates an alert to signal a potential threat. If the data is identified as known attack data, it undergoes further classification. Depending on the classification results, if appropriate measures are available to mitigate the threat, the system will handle the attack without raising an alert. However, if no predefined countermeasures exist, an alert will be triggered to notify the relevant authorities.The framework is shown in Fig. 1. One of the key advantages of using the Gaussian mixture model (GMM) in our Intrusion Detection System is its ability to assign probabilities to each data point, facilitating the determination of whether the data are classified as known attack data, unknown attack data, or normal data. Specifically, if the probability associated with the known attack component is higher, the data is labeled as known attack data. Conversely, if the probability is higher in the normal component, the data is labeled as normal. In cases where both the known attack and normal probabilities are low, the data is classified as unknown attack data. In addition, GMM offers the benefit of being computationally efficient, making it a practical choice for real-time intrusion detection scenarios.

III. Results and Discussion

A. Dataset

We evaluated our proposed system on the CICIDS2017 and CICIDS2018 datasets, both individually and in combination. These datasets are well-established benchmarks for intrusion detection. Table III summarizes the attack classes and their occurrences in each dataset. In our evaluation, we used 10% of each dataset individually and 5% for combined analysis. CICIDS2017 was used for training and CICIDS2018 for testing, leveraging their similarities to achieve generalization. The evaluation results will be discussed in the results section.

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TABLE III: Attack classes and their number of occurrences in CICIDS2017 and CICIDS2018

Attack class	Details	Count	
		CICIDS2017	CICIDS2018
Benign	-	2,271,320	7,364,941
DDOS	Hulk	230,124	145,199
	DDOS	128,025	775,955
	GoldenEye	10,293	41,406
	DoS slowloris	5,796	9,908
	Slowhttptest	5,499	43
Port scan	-	158,804	-
Brute force	FTP-Patator	7,935	53
	SSH-Patator	5,897	94,041
Web-attack	Brute Force	1,507	555
	XSS	652	227
	SQL Injection	21	79
Botnet	-	1,956	144,535
Infiltration	-	36	129,786
Heartbleed	-	11	-

B. Experimental Results

During implementation, the Fisher score method was used to select the top 30 features from each dataset after standard scaling. Gaussian Mixture Models (GMM) and Principal Component Analysis (PCA) were then applied for clustering, separating benign and attack instances. Table IV shows the performance of the GMM model with various PCA components, where the highest accuracy is 81% for 7 PCA components on the CICIDS2017 dataset and 75% for 2 PCA components on the CICIDS2018 dataset. For classification, we used Neural Networks (NN), Naive Bayes (NB), Decision Trees (DT), K-Nearest Neighbors (KNN), and Logistic Regression (LR) to classify anomalies. DT achieved the highest accuracy (99%) for 8 classes of anomaly on CICIDS2017, while KNN achieved 76% for 6 classes of anomaly on CICIDS2018, as shown in Table V. For generalization, CICIDS2018 was used for anomaly detection with GMM, and CICIDS2017 for classification with the same supervised techniques The GMM model has achieved 75% accuracy, while NN achieved 42% when scaling datasets individually, dropping to 34% when scaled together. Other models performed as follows: NB - 13%, DT - 21%, KNN - 18%, and LR - 15%. The drop accuracy indicates the importance of the data acquisition techniques for different attacks as both datasets are generated in different environments.

IV. Conclusion

An early detection of zero-day attacks is crucial for the network and system safety. However, we have tried to propose a system, using the combination of supervised and unsupervised techniques. We apply the CICIDS2017 dataset which exhibits strong detection and classification capabilities, whereas the CICIDS2018 dataset, with its heavy focus on DDoS attacks, resulted in weaker classification performance, particularly in unsupervised methods. This discrepancy is largely due to the dataset's imbalance and specific emphasis on DDoS attacks, along with variations in data collection practices, which hindered the generalization capabilities of the models trained on these datasets. Our system shows some weakness on the generalization of datasets from different systems generated. Future work should focus on larger, balanced datasets, advanced computing resources, and hyperparameter tuning to enhance detection models' generalization, efficient data analysis, and robustness across diverse network environments.

TABLE IV: Performance of GMM Detection model for Different Numbers of PCA Components

PCA	2	6	7	11	15
Accuracy of CICIDS2017	43%	47%	**81%**	74%	72%
Accuracy of CICIDS2018	**75%**	66%	68%	58%	59%

TABLE V: Performance of Datasets for different Model

		Technique				
Datasets		***NN***	***NB***	***DT***	***KNN***	***LR***
CICIDS 2017	***Accuracy***	97%	68%	**99%**	98%	85%
	Precision	0.97	0.65	0.99	0.97	0.815
	Recall	0.93	0.623	0.978	0.97	0.813
	F1 score	0.95	0.421	0.96	0.98	0.833
CICIDS 2018	***Accuracy***	58%	43%	63%	**76%**	56%
	Precision	0.591	0.451	0.552	0.691	0.631
	Recall	0.444	0.323	0.503	0.622	0.434
	F1 score	0.412	0.266	0.505	0.632	0.405

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